

Deep Learning with Explainable AI for Medical Insurance Claim Fraud Detection

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Abstract—We evaluate five deep tabular architectures under a shared insurance-fraud protocol. The same supplied dataset, split fingerprint, feature registry, and metric canon as a traditional baseline are reused. Three seeds quantify stability, while occlusion, deletion-faithfulness, and jitter-stability make explanations comparable across architectures. The selected deep model is chosen from validation evidence before a locked test evaluation. The small supplied workbook and missing policy/document evidence limit generalization; results are therefore a transparent research extension rather than a deployment claim.

Keywords—deep learning, fraud detection, tabular data, explainable AI, insurance

1. INTRODUCTION

Deep networks can represent nonlinear feature interactions and high-cardinality relations, but tabular deep learning is not automatically superior to boosting. This paper tests the proposition under controlled data and evaluation contracts.

2. RELATED WORK

MLP, attentive tabular networks, convolutional tabular models, autoencoders, and transformers offer different representation biases. Explanation research cautions that native attention or masks are not automatically faithful explanations; we therefore use a common perturbation-based audit.

3. DATA AND COMPARABILITY

The supplied workbook contains 4,500 rows and a six-percent fraud prevalence. Approach 2 verifies input checksum 952b6e23c9845b90... and split fingerprint 3bc2230ff033d29d.... Numeric and one-hot features are transformed using the train-only Approach 1 state.

4. ARCHITECTURES

The MLP is a batch-normalized baseline. The TabNet-style model produces sparse soft masks across decision steps. The 1D CNN probes local neighborhoods in a declared feature order. The autoencoder hybrid combines reconstruction error with a supervised head. The feature-token transformer applies pre-norm self-attention and mean pooling.

5. TRAINING

AdamW, cosine decay, inverse-frequency fraud weighting, gradient clipping, checkpoints, telemetry, and early stopping are common. Seeds 42, 43, and 44 produce mean and standard deviation rows. The reference budget is intentionally laptop-first; a larger search is future work.

6. RESULTS

The selected model is Feature-token transformer with validation PR-AUC 0.9800, validation F2 0.9709, and test F2 0.9756. The complete table contains all five rows and is generated from `evaluation2/leaderboard.csv`.

7. XAI AUDIT

Occlusion importance zeros one transformed feature and measures mean probability change. Faithfulness compares deletion of top-ranked features with random deletion. Stability uses top-10 Jaccard overlap under small jitter. Native masks, tokens, and reconstruction contributions are reported as auxiliary context.

8. FAIRNESS AND ETHICS

Gender, age band, claim type, and employment slices are audited after inference. The model does not establish claimant intent, and a flagged score is a review recommendation. Small slices and uncertain provenance limit fairness claims.

9. DISCUSSION

The deep approach is valuable when it adds stable performance or explanation quality on richer data. On a small structured matrix, additional optimization can cost more than it returns. The correct conclusion is empirical parity or improvement under a fixed protocol, not architecture marketing.

10. LIMITATIONS AND FUTURE WORK

The reference run does not yet use raw categorical embeddings, temporal sequences, graphs, documents, or an external Indian claims sample. Future work should add these components while preserving the split and evaluator.

11. CONCLUSION

A controlled deep-learning extension is now available with five architectures, three seeds, common XAI, calibration, fairness, and artifact-driven reporting. The evidence supports comparison with later agentic document reasoning without overstating what the supplied data can prove.

TABLE I — DEEP VALIDATION LEADERBOARD

Model	PR-AUC	F2	Std
dl_e_transformer	0.9800	0.9709	0.0058
dl_b_tabnet	0.9782	0.9709	0.0038
dl_c_cnn1d	0.9763	0.9570	0.0033
dl_a_mlp	0.9748	0.9662	0.0047
dl_d_autoencoder	0.9528	0.9362	0.0210

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REFERENCES

1. R. J. Bolton and D. J. Hand, "Statistical fraud detection: A review," **Statistical Science**, vol. 17, no. 3, pp. 235–255, 2002.
2. N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer, "SMOTE: Synthetic minority over-sampling technique," **Journal of Artificial Intelligence Research**, vol. 16, pp. 321–357, 2002.
3. L. Breiman, "Random forests," **Machine Learning**, vol. 45, pp. 5–32, 2001.
4. J. H. Friedman, "Greedy function approximation: A gradient boosting machine," **Annals of Statistics**, vol. 29, no. 5, pp. 1189–1232, 2001.
5. C. Cortes and V. Vapnik, "Support-vector networks," **Machine Learning**, vol. 20, pp. 273–297, 1995.
6. T. Hastie, R. Tibshirani, and J. Friedman, **The Elements of Statistical Learning**, 2nd ed. Springer, 2009.
7. T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in **Proc. KDD**, 2016, pp. 785–794.
8. G. Ke et al., "LightGBM: A highly efficient gradient boosting decision tree," in **NeurIPS**, 2017.
9. L. Prokhorenkova et al., "CatBoost: Unbiased boosting with categorical features," in **NeurIPS**, 2018.
10. S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in **NeurIPS**, 2017.
11. V. Chandola, A. Banerjee, and V. Kumar, "Anomaly detection: A survey," **ACM Computing Surveys**, vol. 41, no. 3, 2009.
12. M. Herland, T. M. Khoshgoftaar, and R. Wald, "A review of data mining using big data in health informatics," **Journal of Big Data**, vol. 1, 2014.
13. H. Joudaki et al., "Using data mining to detect health care fraud and abuse: A review of literature," **Global Journal of Health Science**, vol. 7, no. 1, 2015.
14. R. A. Bauder, T. M. Khoshgoftaar, and N. Seliya, "A survey on the state of healthcare upcoding fraud analysis and detection," **Health Services and Outcomes Research Methodology**, vol. 17, pp. 31–55, 2017.
15. F. Pedregosa et al., "Scikit-learn: Machine learning in Python," **JMLR**, vol. 12, pp. 2825–2830, 2011.
16. P. Virtanen et al., "SciPy 1.0: Fundamental algorithms for scientific computing in Python," **Nature Methods**, vol. 17, pp. 261–272, 2020.
17. W. McKinney, "Data structures for statistical computing in Python," in **Proc. SciPy**, 2010, pp. 56–61.
18. National Health Authority, "Ayushman Bharat Digital Mission," public programme documentation, <https://abdm.gov.in/>.
19. Insurance Regulatory and Development Authority of India, public health-insurance and policyholder-protection resources, <https://irdai.gov.in/>.
20. Ministry of Health and Family Welfare, Government of India, public health statistics and programme resources, <https://mohfw.gov.in/>.
21. Python Software Foundation, "Python documentation," <https://docs.python.org/3/>.
22. NumPy Developers, "NumPy documentation," <https://numpy.org/doc/>.
23. pandas development team, "pandas documentation," <https://pandas.pydata.org/docs/>.
24. scikit-learn developers, "Model evaluation documentation," https://scikit-learn.org/stable/modules/model_evaluation.html.
25. supplied workbook, `Health Insurance Fraud Claims.xlsx`, repository-provided academic data snapshot; provenance and redistribution license require confirmation from the data owner.